Ram Calculus and Polynomial Limit Cycles

M277: Degree-Uniform Claims, Singular-Passage Audit, and Quantitative Return-Panel Certificates

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Abstract

The September 24, 2026 manuscript *Uniform bounds for planar polynomial limit cycles* claims a finite upper bound depending only on the polynomial degree, over all coefficient vectors and the entire plane. That assertion is substantially stronger than finiteness for one field, finite cyclicity in a restricted family, or a certified count on one return section. We examine the complete mathematical source, its packet separation and differential closure, ordinary elimination, finite passage templates, degeneration treatment, and absolute-to-uniform projection argument. We distinguish verified algebraic and geometric interfaces from the extensive analytic chain not independently certified here. The local Ram corpus directly represents the flow, variational equations, return events, and singular transition models. From its finite-space and physical-metric framework we derive explicit moment-to-jet error bounds, validated flow and event estimates, Bernstein–Rolle upper certificates that allow multiple zeros, and robust exact counts on exhaustively covered return intervals. Exact saddle and cubic examples identify the loss of uniform margins at degeneration. Rational hyperbolic witnesses give a cofinal lower-bound search, not an upper-bound stopping rule. These proved local results and conditional interfaces do not constitute a new proof of the degree-uniform assertion, a numerical value of the Hilbert number, or full certification of the source manuscript.

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1 Exact Target and Evidence Standard

1.1 The quantifiers that must not be interchanged

For $d \geq 1$, let $\mathcal{V}_d$ be all vector fields

$$V = (P, Q), \qquad \dot{x} = P(x, y), \quad \dot{y} = Q(x, y), \qquad P, Q \in \mathbb{R}[x, y], \quad \max(\deg P, \deg Q) \leq d.$$

A limit cycle is an isolated nonconstant periodic trajectory, counted as its geometric image, without multiplicity. Equilibria and the nonisolated orbits of a period annulus are not limit cycles. Define

$$H(d) = \sup_{V \in \mathcal{V}_d} \#\{\text{limit cycles of } V\} \quad \text{in } \mathbb{N} \cup \{0, \infty\}. \tag{1}$$

M277, Section 1, `thm:uniform`, asserts

$$\forall d \geq 1 \; \exists B(d) < \infty \; \forall V \in \mathcal{V}_d : \qquad \#\{\text{limit cycles of } V\} \leq B(d). \tag{2}$$

There is no coefficient-size bound, compact phase-space restriction, hyperbolicity assumption, exclusion of degenerate singularities, or prescribed period bound in this statement. The asserted bound is noneffective: the source does not give a computable formula for $B(d)$. Its conclusion is finiteness of the Hilbert number, not its exact value or a classification of cycle arrangements.

The following assertions have different logical content:

1. A fixed V has finitely many cycles. The bound may depend on every coefficient and on its singular geometry.
2. A specified compact coefficient set or degeneration has locally bounded cyclicity, under explicitly uniform hypotheses.
3. A validated first-return map on a covered interval has exactly q simple zeros, or at most q zeros of any finite multiplicity.
4. Every field of degree at most d, everywhere in the plane, obeys one common bound. This is (2).

Neither a finite list of examples nor pointwise finiteness on a compact coefficient sphere implies the fourth assertion. Positive time scaling normalizes nonzero coefficient vectors without changing trajectories, but it does not supply uniform cyclicity at the boundary of regular parameter regimes. The zero field has no nonconstant isolated orbit.

1.2 Applicability and what this report establishes

The catalogue row 277 assigns level 5 applicability. This is justified by direct overlap with the central return-operator and singular-passage problem in local `res-math/TR-item-015-astra.tex`, not by a claim that the Ram corpus already proves (2). Substantial estimates may remain open even when the representation is exactly the right one.

The report uses three evidence categories:

Proved here. Explicit finite-space error identities, residual bounds, event differentiation, zero-count certificates, and exact model calculations, with their hypotheses and proofs below. Classical ingredients are identified; no priority claim is made for Gronwall, Rolle, Bernstein enclosures, or persistence of hyperbolic cycles.

Source claim with inspected argument. Theorems in M277, whose full mathematical derivations were read. Selected interfaces were independently rederived; reading all sections is not verification of all analytic estimates.

Conditional or unchecked. The all-degeneration packet and closure theorems needed for the global conclusion. Neither a Ram coordinate change nor this audit discharges them.

No numerical experiment below is presented as a new polynomial field with many cycles. The exact cubic model is a familiar benchmark. Abstract scalar fixtures are not automatically realizable return maps of polynomial fields.

2 Primary Literature and Status

2.1 What the established and recent sources address

The historical individual-finiteness problem is associated with Dulac, Ilyashenko, and Écalle. It must be separated from uniformity in d. Yeung's published *Dulac's Theorem Revisited* (2025) identifies a problem in an asymptotic nonoscillation argument in Ilyashenko's 1991 approach and restricts the scope of the argument under discussion [4]. This is not a constructed polynomial field with infinitely many cycles, and does not by itself invalidate every independent approach to individual finiteness. M277 explicitly builds a different packet criterion and does not use prior individual finiteness in its last step.

Gasull and Santana's Theorem 2 and Proposition 2 [3] show that the supremal count can be approached through hyperbolic cycles, whether the supremum is finite or infinite. Their statement is for exact degree; applying it to each degree at most d gives the corresponding reduction for (1). It is not a degree-uniform upper bound. This published paper is *Proceedings of the AMS* **153** (2025), 669–677; the 2024 date in the local Ram bibliography refers to its preprint. We use the public preprint's explicit theorem locators as well as the publication metadata.

Recent work also underscores the importance of matching classes. Huzak and Perez (2026) construct arbitrarily many canard cycles in regularized piecewise-linear systems, explicitly not in one fixed-degree polynomial class [8]. This is not a counterexample to (2). Eshkobilov, Kadyrov, and Mamayusupov's April 2026 preprint gives a Chebyshev replication lower bound $H(nm + m - 1) \geq m^2 H(n)$ [9]. Replication increases degree; it does not settle a finite upper bound at fixed degree. These recent results are used for scope and status, not as unexamined proofs imported into the present certificate theorems.

2.2 External theorem interfaces in M277

Lion–Rolin's preparation theorem and Opris's simultaneous preparation formulation apply to specified definable or globally subanalytic data [5, 6]. In particular, Opris, Fact 2.15 and Proposition 2.55, give finite simultaneous preparations with a common center and rational powers in the globally subanalytic case. M277 applies preparation to scalar fields and geometric data before introducing its passage maps. It does *not* get definability of arbitrary polynomial flows for free from preparation. The new packet analysis carries that additional burden. The exact strong-unit and finite-complexity specializations in M277 Section 9 still require their stated hypotheses to hold throughout the recursive construction.

Bierstone–Milman, Corollary 2.7 [7], gives local finiteness of connected components of semianalytic sets. This is an appropriate bounded-corner input after the equations extend analytically across the limiting tuple. It does not say that an arbitrary analytic set on an unbounded open domain has finitely many components. Sard's theorem is used in M277's regular-fiber projection argument; it is not itself an absolute-zero theorem.

2.3 Status as checked on October 7, 2026

M277 is a newly released claim of (2). The public OpenAI repository says the collection contains results at different verification stages and that unformalized results may have issues [2]. The presence of a related family member or a formal artifact for a restricted Liénard problem must not be promoted to formal certification of M277's all-degree theorem. This report did not run or establish a complete formalization of M277.

The recent primary literature located above distinguishes its restricted or lower-bound results from the general uniform problem. Its publication dates mostly precede M277 and therefore cannot decide the new manuscript's correctness. Conversely, a release announcement cannot establish community acceptance. The defensible current statement is: *a degree-uniform proof is claimed in M277; the complete chain has not been independently certified in this report.* No counterexample to (2) is established here.

3 Full-Source Proof Audit

3.1 Reading scope and source coordinates

The manifest identifies the local source with commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a` of the public source [1]. All mathematical sections `01-introduction.tex` through `11-counting.tex`, the four mathematical figures, the main file, macros, and bibliography were read. Vendored packages were excluded. In this section, filenames are relative to the source build directory listed in Appendix A; labels are exact TeX locators. This is the manifest's revision identifier, not a claim of a separately fetched bytewise comparison. The eleven sections comprise 11,936 lines. The audit does not certify every auxiliary bound, external theorem proof, or uniform constant in that source.

3.2 Flags, separation, and ordinary elimination

Sections 2–3: more than a formal expansion. `02-flags.tex`, `def:flags-flag` and `lem:flags-saturation`, organize separated scales $Z_1 \gg \cdots \gg Z_m \gg 1$, with $f_i = \log Z_i$ and $h_i = e^{-Z_i}$. Dependent logarithms have exact affine identities in later scales; ordinary bounded remainders remain arguments. Saturation is chosen after passing to a subsequence. It is not a degree-uniform bound on flag height.

`03-separation.tex`, `def:packet` and `thm:separation`, require actual sectorial functions, two lateral coefficient trees, controlled forward domains, iterated left-finite supports, and quantitative source costs. A zero coefficient tree alone is not the hypothesis. The source treats retained cutoff costs and join costs separately; the weighted estimate `lem:separation-weighted` is used to upgrade arbitrary-order decay to exact vanishing. This is the crucial replacement for the invalid generic inference "flat expansion implies zero."

The audit read the weighted maximum argument, Cauchy correction, side barriers, and zero-subtree upgrade. Its correctness requires estimates on the whole declared forward domains with the indicated independence from the large testing weight, not merely bounds along the original real sequence. These analytic estimates were not all independently reproduced. No demonstrated contradiction to this separation theorem is asserted here.

Section 4: differentiation before tying arguments. `04-calculus.tex`, `ass:calculus-primitive`, `thm:calculus`, and `lem:calculus-refinement`, keep independent kernel inputs, current backgrounds, and a fixed finite coherence order. Allowed operations include finite sums, products, independent derivatives, monomials, regular analytic compositions, and implicit maps whose limiting

zeroth-order Jacobian is invertible. A pointwise invertible graph that degenerates at the limit is *not* covered by that last premise.

For example, changing the tiny background x in $\exp(\exp(S + x))$ can change its normalized value by order one even when $x \to 0$. Freezing backgrounds is not legitimate. The source explicitly avoids it. Likewise, equality of two extensions on an argument graph does not imply equality of their ambient partial derivatives. The proper chain rule is essential in Sections 10–11.

Section 5: actual zeros, not approximate contacts. `05-elimination.tex`, `prop:eliminatio n-recursion`, distinguishes exact, visible, and coarse centers. A derivative-centered expansion can remove the degree-$m-1$ term without removing the constant residual; the prototype $z^2 + c(y)$ makes that distinction unavoidable. Normalized visible center polynomials reduce multiplicity. Negligible centers require a separate comparison floor and replay of the actual root equations. The common-domain argument, `lem:elimination-common-domain`, must fix conditioning and source requirements before choosing arbitrary accuracy orders.

`thm:absolute-zero` has a strong sequential hypothesis: every sequence of solutions in the full tuple must admit either an analytic bounded extension or a saturated, retestable flag with a recoverable old free input. Retestability in `def:elimination-retestable` is invariance of the relevant coefficient germs under allowed tiny changes, not equality of a formal expansion at one anchor. This theorem is an absolute isolated-zero criterion for the specified systems; it cannot be replaced by fixed-parameter finiteness.

3.3 The three passage libraries

Section 6: additive passages. `06-additive.tex`, `thm:additive-packets`, treats the action $A(w) = w + \gamma \log(w/q)$, $\gamma = \delta q$, with $A(p) = Q$ and transfer charge $e^{-(Q-q)}$, in damped and layer-driven models. The vanishing conditions on the nonlinear forcing and smallness of amplitudes are part of the theorem. Formal asymptotics are not the entire output: the argument uses actual complex continuation, Gevrey estimates, fixed-side convergent charge tails, and exponentially small retained errors.

The cutoff retains a finite union of charge strips $\min(a, b_f) \le K$, not an unrestricted convergent mixed series and not just total degree $\le K$. The two-sided representation retains the inner anchor: in a schematic expression $V(p) + P(v_{\text{in}} - V(q))$, a lateral change in V may cancel only if the $V(q)$ contribution is retained. The audit found no warrant for simplifying these anchors away.

Section 7: regular and stable passages. `07-regular.tex`, `thm:regular-packets`, assumes a regular base flow with state room and layer vanishing, or a stable orientation for the damped model. Finite charge expansions and Gevrey truncations have request-dependent constants. Retiring a cutoff at its designated flag level is part of the source accounting. No uniformity in all truncation orders simultaneously is stated or needed.

Section 8: mixed long clocks. `08-mixed.tex`, `thm:mixed-packets`, uses two long clocks L, W and the physical slope $c = W/L$. The layer rates are $\alpha + \beta c$. In the small-slope regime, each forcing monomial must contain a designated fast layer; purely slow rates cannot be silently bounded below uniformly. Comparable, small-ratio, and large-ratio cases are separate. The proof groups colliding residues analytically rather than treating artificial partial-fraction poles as true singularities. Its pole-avoiding columns, fixed-side tails, and cutoff retirement are substantial estimates, not consequences of a finite Gram matrix.

`prop:calculus-library-uniformity` assembles finite-request control on the common forward domains. This is an important interface to verify independently. "Uniform for one fixed finite request" must not become "uniform over every derivative order and every degree."

3.4 Finite geometry and all boundary regimes

Section 9: a finite alphabet before taking sequences. `09-subdivision.tex`, `thm:subdivision-preparation`, prepares the scalar rational fields, poles, and center branches. Numerator balance radii as well as pole radii are needed. Moving centers retain their time-derivative term. A bounded-order derivative argument distinguishes small, large, and comparable center speeds; its recorded complexity decreases even when one step temporarily retains the same order. The special weighted redo is argued to occur at most once on the relevant branch, after the exponent data have been fixed.

`thm:finite-templates` is the indispensable degree-only geometric input. The crossing argument has genuine planar content: a simple periodic Jordan curve has alternating crossings along an arc, whereas a fixed nonzero sign of $\det(V, T)$ fixes their orientation, so such a transversal arc can be crossed at most once. A tangent nonsingular arc lies on one trajectory. Finite subdivisions then control word lengths and exceptional orbits. This does not bound the crossings of an arbitrary nonclosed orbit. The proof's claimed finite preparation and recursive termination must hold uniformly over the entire coefficient family, not just a chosen field.

Section 10: exact ambient kernels. `10-terminal.tex`, `thm:terminal-reduction`, constructs actual ODE transfers on independent-argument domains, with strict margins and larger analytic state boxes. The no-zero, simple-root, and annular alternatives include short clocks, bounded action, one long clock, and two long clocks. Near-matching identities are established before setting the mismatch equal to zero.

For instance, if $z_+ = v_+ - v_- - \kappa S$, then the right-layer amplitude

$$\chi Q^p e^{av_+ - az_+} = \chi Q^p e^{av_- + a\kappa S}$$

does not actually depend on the proposed final state on the tied graph. Differentiating without that cancellation would corrupt the matching rank. The source retains it in `lem:terminal-tied-endpoints` and `lem:terminal-ambient-extension`.

The boundary-corner lemma `lem:terminal-boundary-corners` is separate from positive-margin completeness. The latter says nothing about a sequence whose margin tends to zero. Bounded clocks have normalized fields such as $v_s = -Wv + Lg$, with exponents $\alpha L + \beta W$, analytic even at $(L, W) = (0, 0)$ when all other retained arguments remain in their larger boxes. A ratio is either retained as a graph coordinate, or, if it diverges, moves the enlarged tuple into the large-scale analysis. Analytic extension across these corners is needed for every old independent derivative requested by the closure, not merely along the physical ties.

3.5 Absolute zeros to a uniform fiber count

The exact closure hypothesis. `11-counting.tex`, `def:counting-domain` and `thm:projection-count`, concern $F(p, x) = 0$ on an analytic domain specified by finitely many strict margins $g_j > 0$. For m equations in n fiber variables, $\operatorname{rank} F_x = m$ is required on the zero set. The absolute isolated-zero property must hold for *every* finite system generated by the permitted derivatives, algebra, inverses on their retained nonzero domains, added coordinates, graph ties, multipliers, determinants, and parameter copies or specializations.

Here "absolute" means isolated solutions in the full tuple, including variables previously called parameters. A zero set consisting of many nonisolated curves would not violate that property for the original equations alone. For example, $\sin x = 0$ on $0 < x < p$ has no isolated total zeros but unbounded finite fiber counts. Adding $p = x + 1$ exposes infinitely many isolated total zeros. M277 explicitly notes this sort of obstruction; its closure hypothesis is designed to rule it out.

Why the projection argument is stronger than compactness. The proper barrier

$$\rho(p, x) = 1 + |p|^2 + |x|^2 + \sum_j g_j(p, x)^{-2} \tag{3}$$

uses positive-margin completeness. A fixed linear tilt remains proper and bounded below. In the square case, the proof inducts on parameter dimension, selects a tilt by Sard and Fubini, and bounds its nondegenerate constrained critical points on one-dimensional intermediate fibers. The bordered Jacobian condition is exactly the nonsingularity of the constrained Hessian. Isolated solutions of the resulting square system are controlled by the inductive hypothesis. Persistence of finitely many regular roots transfers the bound from generic to exceptional parameter values. For positive-dimensional regular fibers, a common tilt on a countable collection of witness fibers reduces components to these critical-point systems.

This logical mechanism was traced, including why new determinant margins and multiplier variables remain in the required closure. No specific algebraic defect was found in this reduction under its full hypotheses. It is not a substitute for proving those hypotheses for the passage library. A compactness-only paraphrase would remove the main argument.

Graph lifting and bounded corners. The lemmas `lem:counting-jet-lift`, `lem:counting-cleared-closure`, and `lem:counting-graph-recovery` preserve isolation under exact local graph enlargements. The graph Jacobian need be invertible at each selected point, not uniformly invertible or invertible at its limit. Finite derivatives are rational expressions in analytic jets; denominators can be cleared at the selected points while all original nonzero-domain restrictions are retained. An open restriction cannot turn a nonisolated analytic germ into an isolated germ at an interior point.

For an exact change $\Phi(Q, q) = \Psi(S, M)$ with $S = Q - q$, $M = S/q$, the old partial derivatives are

$$\Phi_Q = \Psi_S + q^{-1}\Psi_M, \qquad \Phi_q = -\Psi_S - (Q/q^2)\Psi_M. \tag{4}$$

These identities were independently checked. Equating Φ_q with one new ambient partial derivative would be wrong. A finite subdivision number chosen once after a subsequence is permissible in an absolute-zero contradiction; a different number at every selected point would not give one fixed finite enlarged system.

The remaining analytic handoff. The completion of `thm:matching-system`, particularly lines 705–934 of `11-counting.tex`, divides enlarged tuples into bounded analytic corners and divergent flag regimes. It asserts that all the primitive constructions and retained graph equations satisfy retestability, and that a recoverable old free input can be varied inside each point's possibly shrinking graph neighborhood. This is precisely where the packet separation, differentiated calculus, boundary-corner library, and ordinary elimination must work together. The report has not supplied an independent proof of that entire assertion. Calling it unverified here does not demonstrate a gap in the source.

3.6 Cycle representations, rotation, and the whole plane

The matching system leaves tangential cut positions free. For a word with k links and n_{aux} graph variables there are $2k + n_{\mathrm{aux}}$ unknowns and $k + n_{\mathrm{aux}}$ equations. At a hyperbolic target, the transverse linearization is

$$\mathrm{d}r_{j+1} - h_j\,\mathrm{d}r_j, \qquad \prod_{j=1}^{k} h_j \neq 1. \tag{5}$$

It has rank k; the fiber has dimension k, not zero. The source uses connected components, not the number of cut placements. On a maximal-minor chart, the analytic first-endpoint map sends a neighborhood of a target into its isolated cycle. The cycle is a compact embedded analytic curve. The local identity theorem makes the interior of its inverse image both open and closed in a connected analytic component. Consequently two distinct cycles cannot have targets in one component. This avoids the incorrect claim that all remote matching points must themselves be physical passages.

The rotation lemma `lem:counting-rotation` uses $V_\mu = V + \mu JV$, $J(a, b) = (-b, a)$. Along a cycle the determinant of the field and the parameter variation obeys

$$w' = (\operatorname{div} V)w + |V|^2, \quad w(0) = 0, \qquad w(T) > 0.$$

Thus the return displacement has $D_\mu \neq 0$. Its zero set is $\mu = \psi(r) = ar^m + O(r^{m+1})$. Odd m yields one nearby simple root for either sign; even m yields two for one sign. For a finite selection, one common sign gives at least its original total number, since

$$\max(O + 2E_+, O + 2E_-) \geq O + E_+ + E_-.$$

It need not preserve every individual even-multiplicity cycle. This argument agrees with the finite-selection hyperbolic reduction used in the public literature; it does not assume individual finiteness.

Conditional on all preceding inputs, the source obtains

$$B_*(d) = T(d) + \sum_{w \in \mathcal{W}_d} \sum_{I} N_{w,I} < \infty, \tag{6}$$

where $T(d)$ counts exceptional geometric cases and I indexes maximal minor charts. Every finite set of cycles can be put in a fixed square by $z = a + Ru$, $R > 0$, since $R^{-1}V(a + Ru)$ still has degree at most d. Rotation keeps the selected perturbed cycles inside that square. If more than $B_*(d)$ cycles existed, choosing a finite larger subset would contradict the bound. This last inference is sound without an individual-finiteness premise, provided the uniform square bound has actually been established.

3.7 Audit verdict and actionable verification work

The source contains explicit defenses against several tempting but incorrect shortcuts: loss of flat terms, frozen backgrounds, differentiation after an unjustified tie, uniformity inferred from fixed-field finiteness, and degeneration ignored because positive margins were once available. The present reading does not establish a concrete fatal error in those defenses. It also does not establish the full proof.

The highest-value independent checks are:

1. Reprove the weighted separation estimate, with every source cost and constant dependence, then test the zero-subtree upgrade after multiplying by earlier growing monomials.

2. Audit the common-domain quantifiers through a finite stack of derivatives and implicit charts. Distinguish fixed coherence order from arbitrary requested accuracy.

3. Check mixed rate collisions and all small/large slope regimes on independent arguments, with retained anchor and cutoff errors.

4. Verify finite template complexity in Section 9, especially moving centers and the one-redo assertion, before using the finite sum (6).

5. Verify all vanishing-margin cases in Section 10 and retestability for the full generated systems in Section 11, including newly unbounded multiplier coordinates. A derivative check only on the physical graph is insufficient.

These are unresolved audit obligations, not a list of demonstrated counterexamples. The local corpus supplies useful certificates for parts of this program, not a replacement for its analytic closure theorem. The independent local sidecar `_audit/M277-source-audit.txt` (independent source review, October 7, 2026), read in full, reaches the same bounded disposition. Its Sections 3–5 independently reconstruct the projection, component-injection, rotation, and graph-lifting implications; its U1–U5 in Section 6 identify the analytic obligations still needing verification. Agreement between bounded audits is not full-chain certification.

One substantive scope correction from that audit is adopted here. TR-item-015, lines 351–360, describes a degree-controlled effective parameterization and numerical-complexity program as what one would still need. Read that as one sufficient constructive strategy, not a necessary condition for a noneffective existence theorem. M277 need not bound all sequence-wise flag heights, internal subdivisions, or numerical stopping times uniformly in d to make its stated projection argument work.

4 A Domain-Correct Ram Representation

4.1 What is imported from the local corpus

TR-item-015 directly addresses validated Volterra flow operators, variational derivatives, first-return maps, singular transition panels, and the uniformity barrier. Its revised counting convention requires one selected oriented intersection per counted geometric cycle, or a proved quotient of multiple intersections. Its September 2026 additions give event differentiation, a scalar multiplier identity, quantitative perturbative counting, and cofinal rational hyperbolic witnesses.

TR248 distinguishes a finite recoverable model from an uncontrolled infinite-dimensional tail. Its `thm:finite`, `thm:perturb`, and physical-coordinate discussion give stable inversion and operator-error accounting [10, 11]. The revised foundation report's `eq:moments` and "A concrete polynomial Ram construction" require full column rank, compatible data, and the correct physical metric. TR250's `thm:transfer` preserves physical norms under consistent invertible coordinates; it does not create a new dynamical zero theorem. Exact local paths and read sections appear in Appendix A.

4.2 Moments, model tails, and derivative norms

Let $I = [a, b]$, $a < b$, and let $\phi_1, \dots, \phi_N \in C^r(I)$ be real linearly independent functions with positive physical Gram matrix

$$G_{ij} = \int_I \phi_i(s)\phi_j(s)\omega(s)\,\mathrm{d}s, \qquad G \succ 0.$$

For ordinary polynomial panels an integrable weight $\omega > 0$ almost everywhere suffices. A finite positive quadrature measure instead needs enough unisolvent nodes. Let ℓ_i be specified finite measurements, such as polynomial/exponential weighted Ram moments, and put $A_{ij} = \ell_i(\phi_j)$. A positive definite measurement metric W defines the deterministic norm used below; it need not be a covariance.

Theorem 4.1 (Physical moment-to-jet certificate)**.** *Suppose the actual displacement has a decomposition* $D(s) = \phi(s)^T c + r(s)$ *with* $\left\|r^{(j)}\right\|_\infty \le \eta_j$ *for* $0 \le j \le r$*. Assume* A *has full column rank* N *and*

$$m = Ac + \delta, \qquad \left\|W^{1/2}\delta\right\|_2 \le \epsilon, \qquad \alpha = \sigma_{\min}(W^{1/2}AG^{-1/2}) > 0.$$

Define $\widehat{c} = (A^TWA)^{-1}A^TWm$ *and* $\widehat{D} = \phi^T\widehat{c}$*. Then*

$$\left\|(D - \widehat{D})^{(j)}\right\|_\infty \le E_j := \eta_j + K_j\epsilon/\alpha, \qquad K_j = \sup_{s\in I}\sqrt{\phi^{(j)}(s)^TG^{-1}\phi^{(j)}(s)}. \tag{7}$$

The error δ *must include measurement of the tail, as well as quadrature and observation errors. A bound on measurement noise alone is not sufficient if* m *measures* D *rather than its finite model.*

Proof. Set $x = G^{1/2}c$ and $T = W^{1/2}AG^{-1/2}$. Weighted least squares gives $\widehat{x} - x = T^\dagger W^{1/2}\delta$, so its norm is at most ϵ/α. Cauchy–Schwarz bounds the jth derivative of $\phi^T(\widehat{c} - c)$ by the displayed evaluation norm times that coefficient error. Add the separate C^j tail bound. □

An invertible change of trial coordinates leaves these physical quantities unchanged when G and A are transformed together. It cannot make an unobserved mode identifiable. For an implemented matrix $\widetilde{T}$ with $\left\|\widetilde{T} - T\right\| \le \zeta$ and a rigorous singular-value lower bound $\underline{s} > \zeta$ for $\widetilde{T}$, one has $\sigma_{\min}(T) \ge \underline{s} - \zeta$. If decoding uses $\widetilde{T}^\dagger$, then instead

$$\|\widehat{x} - x\| \le \frac{\epsilon + \zeta\,\|x\|}{\underline{s}}. \tag{8}$$

Indeed $\widetilde{T}^\dagger Tx - x = \widetilde{T}^\dagger(T - \widetilde{T})x$. A certified amplitude bound for x is needed in (8); operator error cannot be quietly merged with additive noise independent of the signal.

Obtaining a tail. If the normalized section coordinate is $u \in [-1, 1]$ and the true displacement extends holomorphically to $|u| < \rho$, $\rho > 1$, with Taylor coefficients $|a_k| \le M\rho^{-k}$, then its degree-p Taylor panel has

$$\eta_j \le M \sum_{k=\max(p+1,j)}^{\infty} \frac{k!}{(k-j)!}\rho^{-k}. \tag{9}$$

This follows by termwise differentiation and uniform convergence on $[-1, 1]$. Physical derivatives multiply the bound by $(2/(b-a))^j$. The analytic radius and M must be proved for the actual return map, including its return event. They are not consequences of polynomiality of V. A compactly supported smooth window need not be analytic.

4.3 A fully explicit linear-panel margin

On $[0, 1]$, take $\phi = (1, \sqrt{12}(s - 1/2))$, so $G = I$, and use the two normalized interval integrals $\ell_1(f) = \sqrt{2}\int_0^{1/2} f$ and $\ell_2(f) = \sqrt{2}\int_{1/2}^1 f$. Then

$$A = \begin{pmatrix} 1/\sqrt{2} & -\sqrt{6}/4 \\ 1/\sqrt{2} & \sqrt{6}/4 \end{pmatrix}, \qquad \alpha = \sqrt{3}/2, \qquad K_0 = 2, \quad K_1 = \sqrt{12}.$$

Consequently

$$E_0 \le \eta_0 + 4\epsilon/\sqrt{3}, \qquad E_1 \le \eta_1 + 4\epsilon. \tag{10}$$

These are continuum identities, not sampled matrix evidence. Asymmetric intervals of lengths a and $1 - a$ have squared lower margin $3a(1 - a)$, tending to zero as $a \to 0$. Thus pointwise rank does not imply uniform conditioning even in a two-dimensional Ram model.

5 Validated Flow, Events, and Multipliers

5.1 Volterra residuals supply physical error bounds

For a fixed field, or uniformly for a stated coefficient box, the flow satisfies

$$z(t) = z_0 + \int_0^t V(z(u))\,\mathrm{d}u, \qquad \dot{U} = DV(z)U. \tag{11}$$

Ram polynomial or exponential-polynomial panels can represent the approximate flow and variational solution. The representation is useful only with residual and existence enclosures on an actual tube.

Proposition 5.1 (Flow and variational residual bounds)**.** *Let $\widehat{z}$ and $\widehat{U}$ be absolutely continuous and let the exact flow exist on $[0, T_*]$. Suppose the line segments between $z(t)$ and $\widehat{z}(t)$ lie in a set where $\|DV\| \le L$ and $\|D^2V\| \le H$. Define the residuals almost everywhere by*

$$r_z = \dot{\widehat{z}} - V(\widehat{z}), \qquad r_U = \dot{\widehat{U}} - DV(\widehat{z})\widehat{U}.$$

With initial errors $e_0, e_{U,0}$, valid enclosures are

$$e_z(t) = e^{Lt}e_0 + \int_0^t e^{L(t-u)}\,\|r_z(u)\|\,\mathrm{d}u, \tag{12}$$

$$e_U(t) = e^{Lt}e_{U,0} + \int_0^t e^{L(t-u)}\left(\|r_U(u)\| + He_z(u)\left\|\widehat{U}(u)\right\|\right)\mathrm{d}u. \tag{13}$$

Then $\|z - \widehat{z}\| \le e_z$ and $\left\|U - \widehat{U}\right\| \le e_U$.

Proof. Subtract the flow equations and apply the Lipschitz inequality and Gronwall. For the second statement use

$$(U - \widehat{U})' = DV(z)(U - \widehat{U}) + (DV(z) - DV(\widehat{z}))\widehat{U} - r_U.$$

The Hessian bound gives the second forcing term in (13); apply Gronwall again. □

For $\|r_z\| \le \varepsilon_z$ and $L > 0$, $e_z(t) \le e^{Lt}e_0 + \varepsilon_z(e^{Lt} - 1)/L$; at $L = 0$ use $e_0 + t\varepsilon_z$. The premise about existence and containment may be certified by interval Picard iteration or a strict tube bootstrap. It cannot be concluded by evaluating a residual on an unchecked trajectory. Piecewise panels require continuity or explicit propagation of jump errors.

5.2 First return is an event, not a fixed time

Let a smooth section be $h(z) = 0$, parameterized by $z_0(s)$ with coordinate σ. Suppose there is a unique transverse first return $T(s)$ in a validated bracket, all preceding crossings are excluded, and $|\nabla h \cdot V| \ge \tau > 0$ there. For the fixed-time variation $U = D_s\phi_t(z_0(s))$ at the event, implicit differentiation gives

$$T' = -\frac{\nabla h \cdot U}{\nabla h \cdot V}, \qquad \Pi' = D\sigma\left(I - \frac{V\nabla h^T}{\nabla h \cdot V}\right)U. \tag{14}$$

The subtraction removes the flow-direction component. Omitting it gives a fixed-time derivative, not the return derivative.

For a linear section $h(z) = n \cdot z - c$, $|n| = 1$, and an approximate event $\widehat{T}$ in the same bracket, suppose $|h(\widehat{z}(\widehat{T}))| \le e_h$. The mean value theorem along the true trajectory gives

$$|T - \widehat{T}| \le \frac{e_h + e_z(\widehat{T})}{\tau}, \qquad |z(T) - \widehat{z}(\widehat{T})| \le e_z(\widehat{T}) + M\frac{e_h + e_z(\widehat{T})}{\tau}, \tag{15}$$

where $|V| \le M$ on the bracket. A Lipschitz extension of σ converts this to a return-coordinate error. Enclosing (14) with (13) gives a C^1 return bound. Higher jets require differentiated flow and event equations; the C^0 residual alone does not provide them.

Proposition 5.2 (Scalar return multiplier)**.** *Put $v(s) = \det(V(z_0(s)), z_0'(s))$ on a section with consistent orientation and $v \neq 0$. Then*

$$\Pi'(s) = \frac{v(s)}{v(\Pi(s))} \exp\left(\int_0^{T(s)} \operatorname{div} V(z(t;s))\, \mathrm{d}t \right). \tag{16}$$

At a fixed point the velocity ratio is one. Thus the nontrivial Floquet multiplier is positive and is different from one exactly when the displacement derivative $D' = \Pi' - 1$ is nonzero.

Proof. The flow derivative maps the initial field vector to the final field vector. Its determinant is the exponential of the divergence integral by the two-dimensional variational equation. Taking determinants with the section tangent in (14) eliminates the return-time correction, since it is a multiple of V. This yields (16). □

At an enclosed cycle, let $\ell = \int_0^T \operatorname{div} V(z(t))\, \mathrm{d}t$ and let $\widehat{\ell}$ approximate $\int_0^{\widehat{T}} \widehat{a}(t)\, \mathrm{d}t$ with quadrature error at most e_q. Assume $|\widehat{a}(t) - \operatorname{div} V(\widehat{z}(t))| \le e_{\mathrm{div}}(t)$ throughout $[0, \widehat{T}]$ and $|T - \widehat{T}| \le e_T$. Let L_{div} bound the spatial Lipschitz constant of $\operatorname{div} V$ between the two trajectories on their common interval, and let B_{div} bound $|\operatorname{div} V|$ along both the true trajectory on $[0, T]$ and the approximate trajectory on $[0, \widehat{T}]$, including the unmatched tail. With $T_c = \min(T, \widehat{T})$,

$$|\ell - \widehat{\ell}| \le e_q + L_{\mathrm{div}} \int_0^{T_c} e_z(t)\, \mathrm{d}t + \int_0^{\widehat{T}} e_{\mathrm{div}}(t)\, \mathrm{d}t + B_{\mathrm{div}} e_T. \tag{17}$$

To prove the bound, first remove the quadrature and evaluator errors on the entire computed interval. Compare the exact divergence along the two trajectories on $[0, T_c]$, then bound the remaining true or approximate tail by $B_{\mathrm{div}}|T - \widehat{T}|$. Evaluation error confined to $(T, \widehat{T}]$ when $\widehat{T} > T$ would be missed by integrating e_{div} only to T_c. An interval for the logarithm excluding zero certifies hyperbolicity and its sign gives repulsion or attraction. The initial cycle-location error must already be included in e_z. Away from a fixed point the velocity ratio in (16) must also be enclosed.

6 Finite Certificates for Covered Return Maps

6.1 The coverage contract

All cycle-count statements in this section assume a declared compact section interval I, a total transverse first return on a neighborhood of I, certified existence and first-event selection, and a real displacement $D = \Pi - \mathrm{id}$. Every geometric cycle claimed to be counted must correspond to a fixed point in I, and different counted cycles to different such points. For an *exact* count require a bijection, for example one selected oriented intersection per cycle. A first-return convention alone on an arbitrary disconnected or extended section does not guarantee this bijection: a cycle can have several section intersections and give a periodic orbit of the return map. Equilibria, tangencies, and excluded return domains are not covered.

Theorem 6.1 (Ram–Bernstein–Rolle upper certificate)**.** *Cover I by finitely many closed intervals I_j. On each choose an integer $m_j \ge 0$, a polynomial panel p_j, and a proved bound*

$$\left\| D^{(m_j)} - p_j^{(m_j)} \right\|_{L^\infty(I_j)} \le E_{j,m_j}.$$

If $m_j = 0$, suppose p_j has one sign with magnitude strictly greater than $E_{j,0}$ throughout I_j. If $m_j \geq 1$, impose the same condition on $p_j^{(m_j)}$. Then D has at most

$$N_I = \sum_j m_j \tag{18}$$

distinct zeros in I, and hence at most N_I represented cycles under the coverage contract. Multiple zeros are permitted. The bound is uniform on a coefficient box if the cover, derivative estimates, and return contract hold uniformly on that entire box.

Proof. For $m_j = 0$ the true function is zero-free. Otherwise its m_jth derivative has a strict constant sign. If it had $m_j + 1$ distinct zeros in I_j, repeated Rolle's theorem would give a zero of that derivative, a contradiction. Summing over the cover is conservative even when endpoints or overlap zeros are counted twice. □

For an explicit sign test, write the normalized polynomial $q(t) = \sum_{k=0}^{n} a_k t^k$, $0 \leq t \leq 1$, in Bernstein form. Its coefficients are

$$\beta_i = \sum_{k=0}^{i} a_k \frac{\binom{i}{k}}{\binom{n}{k}}, \qquad \min_i \beta_i \leq q(t) \leq \max_i \beta_i. \tag{19}$$

Nonnegative basis weights summing to one prove the enclosure. Thus $\min \beta_i > E$ or $\max \beta_i < -E$ supplies the required sign. Use the polynomial representing the *physical* derivative, or include the interval-length factor in both polynomial and error. Exact rational subdivision tightens the enclosure without changing the polynomial. An inconclusive Bernstein coefficient is not evidence of an actual zero. These facts agree with the exact conversion and convex-hull results in the local volume 16 chapter.

This theorem adds an upper-count option when a simple-root certificate fails. It does not claim that a finite m_j or a finite cover can be chosen with degree-only complexity through every singular degeneration. The polynomial panel degree and differentiation order are not the degree d of the original vector field.

Theorem 6.2 (Robust exact simple count)**.** *Suppose $\|D - p\|_\infty \leq E_0$ and $\|D' - p'\|_\infty \leq E_1$ on I. Choose disjoint closed intervals $J_1, \ldots, J_q \subset I$ such that:*

1. *p has opposite endpoint signs on each J_i, each endpoint magnitude exceeding E_0;*
2. *on each J_i, p' has one sign and $|p'| > E_1$;*
3. *on $I \setminus \bigcup_i \operatorname{int} J_i$, $|p| > E_0$.*

Then D has exactly one simple zero in each J_i and no other zero in I. Under the bijective coverage contract these are exactly q hyperbolic cycles in the declared class.

Proof. The true endpoint signs persist, giving existence by continuity. The derivative retains a strict sign, giving uniqueness and simplicity on each interval. The complement is zero-free. Proposition 5.2 identifies simplicity with hyperbolicity in the planar return setting. □

Substitute Theorem 4.1 to make E_0, E_1 explicit in the moment error, frame margin, evaluation norms, and actual C^1 tails. Alternatively, interval Newton can supply existence and uniqueness, but its derivative denominator must exclude zero and its inclusion must be strictly inside the root interval. Neither test supplies global coverage.

6.2 Two exact scalar fixtures

With the two-moment model (10), let the decoded panel be $p(s) = s - 1/2$, let $\epsilon \le 1/128$, and let $\eta_0, \eta_1 \le 1/64$. Then

$$E_0 \le \frac{1}{64} + \frac{1}{32\sqrt{3}} < \frac{1}{8}, \qquad E_1 \le \frac{3}{64} < 1.$$

Taking $J = [3/8, 5/8]$ proves exactly one simple zero on $[0, 1]$ for every actual scalar D satisfying those data. This is a reproducible deterministic implication; it is not a new polynomial-flow example until the return and tail premises are supplied.

For $p(s) = (s - 1/2)^2$, an error bound $E_2 < 2$ proves at most two distinct zeros by Theorem 6.1. Small C^0 perturbations $p + \varepsilon$, p, and $p - \varepsilon$ give zero, one, and two zeros near the center, respectively. Thus an upper certificate through a multiple root should not be mislabeled an exact count or a hyperbolicity certificate.

6.3 A coefficient-neighborhood refinement

Suppose a reference displacement D_{θ_0} satisfies the exact count pattern above with endpoint magnitudes at least $b > 0$, complement gap $g > 0$, and derivative margin $\kappa > 0$ on the root intervals. If the same return contract holds on $\|\theta - \theta_0\| \le \delta$ and verified parameter bounds give

$$\|D_\theta - D_{\theta_0}\|_\infty \le K_0\delta, \qquad \|D'_\theta - D'_{\theta_0}\|_\infty \le K_1\delta,$$

then the sufficient conditions

$$K_0\delta < \min(b, g), \qquad K_1\delta < \kappa \tag{20}$$

preserve the exact count on the covered class. This follows directly from sign and derivative persistence, not from a statistical fit. Parameter-sensitivity flow equations, event derivatives, and Ram moment-to-jet estimates can bound K_0, K_1. The return-domain and coverage premises must also persist; a coefficient ball alone does not exclude cycles elsewhere in the plane.

For a near-Hamiltonian displacement $\varepsilon^{-r}D = M + \varepsilon R$, $\varepsilon \ne 0$, the same argument yields $|\varepsilon|B_0 < \min(b, g)$ and $|\varepsilon|B_1 < \kappa$ when $\|R\|_\infty \le B_0$, $\|\partial_s R\|_\infty \le B_1$. This is TR-item-015's perturbative threshold in the same error language. A C^0 remainder alone cannot exclude extra oscillations near a zero.

7 Singular Panels and Explicit Degeneration Limits

7.1 A saddle transition represented without a false Taylor premise

Consider the exact linear saddle

$$\dot{x} = x, \qquad \dot{y} = -\lambda y, \qquad \lambda > 0. \tag{21}$$

From the entry point $(r, 1)$, $0 < r < 1$, to the exit section $x = 1$,

$$T = \log(1/r), \qquad \mathcal{D}_\lambda(r) = r^\lambda = e^{-W}, \qquad W = \lambda T. \tag{22}$$

This follows by solving the two scalar equations. The transit time diverges as $r \downarrow 0$ although the transition value is explicit.

Proposition 7.1 (Exact singular moments and weighted jets)**.** *For real j and $0 < a < b \le 1$,*

$$\int_a^b r^j \mathcal{D}_\lambda(r)\,\mathrm{d}r = \frac{b^{j+\lambda+1} - a^{j+\lambda+1}}{j + \lambda + 1} \tag{23}$$

when $j+\lambda+1\neq 0$, with value $\log(b/a)$ at the removable zero of that denominator. On $(0,1)$ the integral exists exactly when $j+\lambda>-1$ and equals $(j+\lambda+1)^{-1}$. For integer $k\geq 0$,

$$r^k\partial_r^k\mathcal{D}_\lambda(r)=\lambda(\lambda-1)\cdots(\lambda-k+1)\mathcal{D}_\lambda(r), \tag{24}$$

where the empty product is one.

Proof. Integrate the power on a positive interval, taking its logarithmic limit when the exponent is -1. The integral at zero is the ordinary power integrability test. Repeated differentiation proves (24). □

This is a concrete Ram moment panel in a singular power basis, equivalently an exponential panel in T. Parameter derivatives introduce powers of $\log r$ and are also represented exactly on positive compact panels. For example, for $j+\lambda>-1$,

$$\int_0^1 r^{j+\lambda}(\log r)^k\,\mathrm{d}r=\frac{(-1)^k k!}{(j+\lambda+1)^{k+1}}.$$

Differentiation under the integral is justified locally in a parameter range separated from the integrability boundary.

The unweighted derivative $\lambda r^{\lambda-1}$ is unbounded as $r\downarrow 0$ when $0<\lambda<1$. A regular Taylor panel with a uniform C^1 tail at the separatrix is therefore impossible in this coordinate, even in this elementary example. Weighted jets can remain bounded but change the norm and the allowable zero-count argument. Moreover, $r=e^{-1/\lambda}$, $e^{-1/\sqrt{\lambda}}$, and e^{-1/λ^2} all tend to zero as $\lambda\downarrow 0$, while r^λ tends to e^{-1}, 1, and 0, respectively. There is no unique joint limit at that corner. The coupled clock $W=\lambda T$ must be retained. This is a direct model for the source's mixed-clock bookkeeping, not a proof of its nonlinear mixed-packet theorem.

7.2 Flat error can hide infinitely many zeros

On $s>0$, the smooth function

$$f(s)=e^{-1/s^2}\sin(1/s)$$

extends by zero at $s=0$ with every derivative zero there, but has infinitely many positive zeros accumulating at zero. Any finite moment bank or formal zero Taylor tree, without a class and a remainder theorem, cannot exclude this behavior. The example is *not* asserted to be a transition map in M277's packet library. Its role is to show why a Ram approximation cannot replace the source's genuine separation and quasianalyticity obligations. M277 imposes additional lateral and forward-domain hypotheses expressly absent from this example.

7.3 Fixed-degree loss of all simple-root margins

Proposition 7.2 (Cubic degeneration benchmark)**.** *For $\varepsilon\geq 0$, consider*

$$\dot{x}=x(\varepsilon-x^2-y^2)-y,\qquad \dot{y}=y(\varepsilon-x^2-y^2)+x. \tag{25}$$

If $\varepsilon>0$, the only limit cycle is $r=\sqrt{\varepsilon}$, with period 2π and multiplier $e^{-4\pi\varepsilon}$. If $\varepsilon=0$, there is no nonconstant periodic orbit. For a positive radial initial coordinate s, the first return is

$$\Pi_\varepsilon(s)=\left[e^{-4\pi\varepsilon}s^{-2}+\frac{1-e^{-4\pi\varepsilon}}{\varepsilon}\right]^{-1/2}\quad(\varepsilon>0),\qquad \Pi_0(s)=\frac{s}{\sqrt{1+4\pi s^2}}. \tag{26}$$

Proof. Polar coordinates give $r' = r(\varepsilon - r^2)$ and $\theta' = 1$. Every positive radius other than $\sqrt{\varepsilon}$ is strictly monotone, excluding every other periodic orbit. At zero parameter all positive radii decrease. Setting $u = r^{-2}$ yields $u' = -2\varepsilon u + 2$, whose value at 2π gives (26) and its zero-parameter limit. The divergence is $2\varepsilon - 4r^2$, hence is -2ε on the cycle; Proposition 5.2 gives the multiplier. □

Thus

$$|D'_\varepsilon(\sqrt{\varepsilon})| = 1 - e^{-4\pi\varepsilon} \longrightarrow 0. \tag{27}$$

The coefficients stay bounded, the field remains nonzero at the limit, and degree stays three. On the section $y = 0$, the crossing speed at the cycle is $Q(\sqrt{\varepsilon}, 0) = \sqrt{\varepsilon}$, also tending to zero. Positive time normalization of coefficients does not cure the vanishing multiplier gap. The example disproves a degree-only positive lower bound on these certificate margins, not a degree-only upper bound on the *number* of cycles. A proof of (2) may be noneffective and need no such quantitative margin.

8 Composition, Exclusion, and Global Interfaces

8.1 Error propagation across a fixed passage word

Let f_j be exact scalar transfers and $\widehat{f}_j$ approximations on intervals that contain both exact and approximate propagated states. Suppose $\left\| f_j - \widehat{f}_j \right\|_\infty \le e_j$ and the exact f_j is Lipschitz with constant L_j. If E_0 bounds the initial error, recursive subtraction gives

$$E_k \le E_0 \prod_{j=1}^{k} L_j + \sum_{j=1}^{k} e_j \prod_{i=j+1}^{k} L_i. \tag{28}$$

Indeed the next error is at most $L_j E_{j-1} + e_j$. If derivatives are required, differentiate the full compositions and bound the evaluation shifts of f'_j as well as derivative approximation errors; products of nominal derivative values alone are not sufficient. An interval implementation may propagate values and jets jointly.

Equation (28) is useful for one prepared word. It can allocate panel tolerances by downstream amplification instead of spending equal precision at every passage. It is still coefficient- and word-dependent. Large L_j or singular norm changes may make the bound ineffective. No FFT diagonalization of a general return operator is used.

8.2 A quantitative cyclic matching certificate

The transverse rows in (5) admit an explicit inverse bound. This is a concrete refinement of the source's local rank test, useful after the event-corrected holonomy derivatives have been enclosed. It does not certify the full matching domain or its degeneration closure.

Proposition 8.1 (Cyclic inverse margin)**.** *Let $C_h : \mathbb{R}^k \to \mathbb{R}^k$ be given by $(C_h x)_j = x_{j+1} - h_j x_j$, with $x_{k+1} = x_1$. Suppose $|h_j| \le L_j$ and $|1 - \prod_{j=1}^{k} h_j| \ge \gamma > 0$. Empty products below are one and empty sums zero. Define*

$$S_L = \sum_{q=1}^{k} \prod_{\ell=q+1}^{k} L_\ell, \tag{29}$$

$$B_L = \max_{1 \le j \le k} \left\{ \frac{S_L}{\gamma} \prod_{\ell=1}^{j-1} L_\ell + \sum_{q=1}^{j-1} \prod_{\ell=q+1}^{j-1} L_\ell \right\}. \tag{30}$$

Then C_h is invertible and $\left\|C_h^{-1}\right\|_{\infty\to\infty} \le B_L$. If a further matrix perturbation satisfies $\|\Delta\|_{\infty\to\infty} \le \delta < 1/B_L$, then

$$\left\|(C_h+\Delta)^{-1}\right\|_{\infty\to\infty} \le \frac{B_L}{1-B_L\delta}. \tag{31}$$

Proof. For $C_h x = r$, solve the scalar recurrence around the cycle:

$$\left(1-\prod_{\ell=1}^{k} h_\ell\right) x_1 = \sum_{q=1}^{k}\left(\prod_{\ell=q+1}^{k} h_\ell\right) r_q.$$

This uniquely gives x_1, with $|x_1| \le S_L \|r\|_\infty/\gamma$. The remaining recurrence gives

$$x_j = \left(\prod_{\ell=1}^{j-1} h_\ell\right) x_1 + \sum_{q=1}^{j-1}\left(\prod_{\ell=q+1}^{j-1} h_\ell\right) r_q.$$

Taking absolute values proves (30). Finally factor $C_h+\Delta = C_h(I+C_h^{-1}\Delta)$ and use the convergent Neumann series, since $B_L\delta < 1$. □

For example, $k=2$, $h_1=h_2=1/2$ gives $\gamma = 3/4$, $S_L = 3/2$, and $B_L = 2$. Direct inversion of $\left(\begin{smallmatrix} -1/2 & 1 \\ 1 & -1/2 \end{smallmatrix}\right)$ has maximum absolute row sum two, so the bound is exact in this fixture. If $k=1$, it reduces to the scalar inverse gap $|1-h_1|^{-1} \le \gamma^{-1}$.

The norms here are in declared physical transverse coordinates. If Ram moments encode x through an invertible synthesis $x = Rm$, transport the norm as $\|m\|_{\text{phys}} = \|Rm\|_\infty$ and retain the same row scaling. A small raw coefficient-matrix condition number is not a substitute for (30). In the full matching system, auxiliary graph blocks also need their own inverse margins. The free tangential directions remain free; this proposition bounds only the transverse block and does not falsely make the whole fiber isolated. As the return multiplier approaches one, γ can vanish, exactly as in Proposition 7.2.

8.3 Negative certificates with an exact domain

On a simply connected open region U, let $B \in C^1(U)$ and let $V \in C^1(U)$ satisfy $\operatorname{div}(BV) > 0$ everywhere, or the reverse strict sign. No periodic trajectory contained in U can bound a disk there: Green's theorem gives zero boundary flux along a trajectory and a strictly signed interior divergence integral, a contradiction. A simple closed curve in a simply connected planar U has its interior in U, so this excludes every such periodic orbit.

Polynomial Ram moments can represent a candidate B and Bernstein or interval enclosures can certify the divergence sign on a complete cover. For rational B, all denominator zeros must be excluded. On a multiply connected domain the disk may leave the domain, so the stated test no longer applies. A singular multiplier or a sampled sign plot is not a certificate. This supplies genuine exclusion data complementary to positive return-map certificates, without overclaiming global coverage.

8.4 A conditional finite-atlas upper bound

Suppose, for a given d, an independently proved finite atlas covers every degree-at-most-d cycle, apart from at most E_d exceptional cycles per field. Suppose its finitely many return domains carry the coverage contract and certificates of Theorem 6.1, with a sum of orders bounded by a constant C_d over all allowed coefficients. Then $H(d) \le E_d + C_d$. To prove this, assign each nonexceptional cycle one represented zero; summing the certified domain bounds is conservative if a cycle has several representations.

The hypotheses, not the addition, contain the hard theorem. Neither TR-item-015 nor TR248 proves such an atlas and certificate-order bound. M277 takes a different route: a finite geometric atlas followed by absolute-zero closure and uniform bounds on analytic components, rather than a degree-uniform numerical Rolle cover. These routes should not be conflated. In particular, sequence-dependent flags can be compatible with M277's noneffective strategy even though they provide no terminating numerical atlas construction.

9 Cofinal Rational Certificates and Their Stopping Limit

Theorem 9.1 (Lower-bound witnesses suffice for the supremum)**.** *Let $H_{\mathrm{cert}}(d)$ be the supremum of counts of distinct hyperbolic cycles certified in finitely many disjoint local tubes for rational-coefficient fields in $\mathcal{V}_d$. Permit convergent validated flow and event calculations with rational interval data. Then*

$$H_{\mathrm{cert}}(d) = H(d), \tag{32}$$

using the finite-selection hyperbolic reduction described in Section 3.6. No assertion of global completeness for an individual certificate is needed for this lower-bound equality.

Proof. Every certified cycle is an actual cycle, giving the easy inequality. Select any finite set of k cycles of any real field of degree at most d. The rotation argument gives at least k hyperbolic cycles of another field of the same degree bound, in disjoint compact tubes. Their local return zeros are simple. Continuous dependence and the implicit function theorem preserve these cycles in an open coefficient neighborhood. Rational coefficient vectors are dense in that finite space, so a rational field retains all k cycles.

For this one field choose sufficiently small transverse sections and finite return-time brackets. Compact tubes, strict transversality, simple-root derivative gaps, and opposite endpoint signs give positive margins. Rational interval enclosures for polynomial flows, their variational equations, and events converge on these compact time intervals; refinement therefore eventually validates those strict tests. Rational enclosing boxes and sufficiently accurate polynomial panels exist. The lower-bound certificate needs only the finitely many disjoint cycle tubes, not exclusion of every other cycle of the field. Since the finite selection was arbitrary, the reverse supremum inequality follows, including when $H(d) = \infty$. □

The hyperbolic reduction is also supplied by Gasull–Santana; the rational-certificate consequence is developed in TR-item-015. The proof here makes explicit which finite-time margins make a search complete for witnesses. It is not a new proof of finite $H(d)$.

Enumerating rational fields, rational tube data, and finite validation attempts in a dovetailed search gives a nondecreasing integer sequence of certified lower bounds with supremum $H(d)$. If $H(d) < \infty$, it eventually reaches that value, because an integer supremum is attained by some finite witness count. The search has no general certificate that its present record is the last improvement. Thus (32) is lower semicomputability, not a computable upper bound, a complexity estimate depending only on d, or a decision procedure for termination of the global search.

10 Advantages, Costs, and Research Priorities

10.1 Concrete benefits

The Ram framework is directly useful at the operator level: it stores flow and variational panels, represents singular powers and logarithms, and carries physically scaled measurement errors into the derivative norms that counting actually uses. Theorem 4.1 prevents finite-rank evidence from being

silently promoted to a C^1 theorem. Theorem 6.1 gives upper bounds at finite-order multiple contacts, while Theorem 6.2 supplies exact simple counts. Equation (20) reuses a certificate across a verified coefficient neighborhood. Event and scalar divergence formulas make transversality and hyperbolicity explicit. Singular coordinates (22) preserve coupled scales that ordinary Taylor coordinates can obscure.

10.2 Costs and failure modes

The analysis pays for physical containment, first-return selection, tail bounds in the needed derivative norm, interval wrapping, small singular values, near-unit multipliers, long times, and complete coverage. Increasing panel degree does not solve an omitted singular boundary. Adding moments does not remove an unproved analyticity radius. Sampling coefficients does not prove a universal coefficient statement. Probabilistic calibration, if used to choose a coefficient set, does not replace deterministic universal bounds on that set.

Different uses of "finite" must remain separate: finite basis size, finite word alphabet, finite subdivision along one sequence, finite zeros for one system, and a uniform bound for every parameter fiber. Only the last two, under the source's closure assumptions, drive its global counting theorem. Our cubic example shows that uniform numerical margins are stronger than a finite-count assertion and can fail even when the count is identically one.

10.3 Prioritized next steps

1. Build a small certificate checker accepting rational interval data, physical Gram bounds, tail norms, event brackets, and explicit coverage obligations. Its outputs should distinguish exact count, upper count, lower count, exclusion, and unresolved domain.

2. Test the complete flow/event/jet pipeline on (26) as $\varepsilon \downarrow 0$, reporting shrinking gaps and failed cells rather than inventing uniform success.

3. Extend singular Ram panels from the exact saddle to nonlinear regular-singular transitions with sectorial remainder bounds. Keep the coupled clocks and independent-argument differentiation visible.

4. Verify a nontrivial restricted passage family through the entire closure needed by the projection theorem, including derivatives, inverse margins, and multiplier systems. A theorem on values alone would not meet this target.

5. Audit M277's all-regime separation and retestability independently, then the degree-only finite geometry. Only a genuinely established chain would justify upgrading its global claim to a certified result.

11 Conclusion

M277's claimed theorem gives, for each fixed degree bound, a bound uniform over all coefficient vectors and the entire plane, not a fixed-field counting result. Its proof has substantial explicit machinery aimed at precisely that quantifier: actual lateral packet separation, differentiated closure, ordinary elimination, finite geometric templates, and an absolute-to-uniform component argument. The detailed source reading and the checks recorded here do not amount to independent certification of that full chain.

The Ram corpus is nevertheless directly relevant to the core operator model. The proved refinements in this report convert correct physical moments and actual tails into useful flow, event, zero-count, and

parameter-neighborhood certificates. They preserve singular scales and expose degeneration rather than hiding it behind invertible matrices. They establish meaningful local and finite-family conclusions, with an explicit boundary: no new $B(d)$, no new value of $H(d)$, and no global solution claim follows from these certificates alone.

A Exact Evidence and Reproducibility Locators

A.1 Public manuscript actually read

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C:/Users/mural/OneDrive/Documents/OpenAI-722-math
```

The manifest row 277 in `_sources/manuscripts.json` and row 277 in `00-Applicability-Catalog ue/applicability.json` were read. The full mathematical build root is

```
_sources/openai-math/preprints/uniform-bounds-for-planar-polynomial-limit-cycle
s-September-24-2026/build
```

Actual files read: `main.tex`, `macros.tex`, `references/references.bib`; every file in the following list:

```
sections/01-introduction.tex
sections/02-flags.tex
sections/03-separation.tex
sections/04-calculus.tex
sections/05-elimination.tex
sections/06-additive.tex
sections/07-regular.tex
sections/08-mixed.tex
sections/09-subdivision.tex
sections/10-terminal.tex
sections/11-counting.tex
```

The four figure sources read are `figures/coupled-bands.tex`, `figures/matching-arguments.tex`, `figures/proof-roadmap.tex`, and `figures/retained-charges.tex`. All mathematical sections were read, not only the introduction and counting conclusion. Source labels in Section 3 provide exact theorem-level locators; they are not cross-references to the numbering of this report.

A.2 Local Ram sources actually consulted

1. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-ite m-015-astra.tex`. Full report read, including `thm:return`; "Singular Panels and the Global Uniformity Barrier"; "Return-event derivatives and multiplier error," `eq:event-derivative`, `eq:poincare-det`, `eq:multiplier-error`; "An explicit perturbative cycle threshold," `eq: melnikov-threshold`; and "Hyperbolic rational witnesses are cofinal for the Hilbert number," `eq:cofinal`. The real-algebraic topology part is not used as a dynamical counting theorem.

2. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/TechReps/FinalTechReps /TRs/TR248-Stable-Adaptive-Ram-Frames/TR248-Stable-Adaptive-Ram-Frames.tex`. Read "Spaces, measurements, and the exactness boundary," E1, E2 through the physical coordinates and noise discussion, and the appendix "Moment assembly, analytic tails, and exact certificates." Key labels: `thm:finite`, `eq:modelerror`, `thm:perturb`, `eq:decodepert`, `eq:momentgram`. Unrelated model-selection and quantization sections are not claimed as audited inputs.

3. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Read "What a Ram representation means in this report," `eq:moments`, and "A concrete polynomial Ram construction." These explicitly require the selected physical space, full column rank for rectangular banks, a stated off-range projection metric, and a separate tail.

4. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/TechReps/FinalTechReps/TRs/TR250-Compatible-Variational-Ram-Panels/TR250-Compatible-Variational-Ram-Panels.tex`. Read "Spaces, operator domains, and Ram coordinates" and E1 through the discussion of bounded projections. Key label: `thm:transfer`. No claim about the full PDE report is needed.

5. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`. Read "Polynomial enclosures in Bernstein form" and its subdivision subsection. Exact labels: `v16c05:powerbernstein`, `v16c05:conversionproof`, `v16c05:hull`. Only the polynomial enclosure algebra, not the chapter's geometric application, is transferred here.

The independent local sidecar `_audit/M277-source-audit.txt` was also read in full and incorporated as described in Section 3.7. It is not substituted for the source sections themselves. The corpus maps were used for navigation; the claims above are tied to actual source text. This report does not infer new zero theorems from generic Gram algebra or an unspecified Ram transform.

A.3 Verification boundaries

The symbolic derivations in Sections 4–9 are supplied in the report itself. Exact rational inequalities can check the two scalar fixtures. No floating-point experiment is used to certify an unknown polynomial field, no full source formalization was run, and no exhaustive phase-space or coefficient-space atlas was constructed. Public searches used public mathematical titles and public primary pages only; private corpus text was not submitted as a search query. Original sources remain unchanged.

References

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[3] A. Gasull and P. Santana, A note on Hilbert 16th Problem, *Proceedings of the American Mathematical Society* **153** (2025), 669–677. Public preprint: Theorem 2, Proposition 2, and the rotation argument in Section 2. `https://arxiv.org/html/2407.13465v2`.

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[9] O. Eshkobilov, S. Kadyrov, and Kh. Mamayusupov, Limit-Cycle Replication via Chebyshev Pullbacks and a Quadratic Ceiling for Separable Schemes, preprint, April 14, 2026. `https://arxiv.org/abs/2604.12883`.

[10] M. Subbarao, *Hilbert's Sixteenth Problem through Ram Calculus: Certified Real-Curve Topology, Validated Return Operators, and the Remaining Uniformity Barrier*, TR-item-015-astra, local technical report, September 2026 revision. Exact file and sections in Appendix A.

[11] M. Subbarao, *Stable Adaptive Ram Frames and Task-Aware Approximation*, TR248, local technical report, 2026. Exact file and sections in Appendix A.

[12] M. Subbarao, *Compatible Variational Ram Panels*, TR250 and revised foundation version `TR-rev-1-old-tr250`, local technical reports, 2026. Exact files and sections in Appendix A.

[13] M. Subbarao, Ram Calculus book corpus, volume 16, `chapters/ch05d-clearance.tex`, polynomial Bernstein enclosures and exact subdivision. Exact file in Appendix A.